2024-25 First Semester
MATH1053 Linear Algebra I

Assignment 10

Due Date: Due on 10/Dec/2024 (Tuesday), at the end of lecture.

- Write down your **CHN name** and **student ID**. Write neatly on **A4-sized** paper (*staple if necessary*).
- **Late submissions or answers without steps will not be graded.**

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1. Show that if A and B are $n \times n$ matrices and $N(A - B) = \mathbb{R}^n$, then $A = B$.

2. Let $\mathbf{x}$ and $\mathbf{y}$ be nonzero vectors in $\mathbb{R}^m$ and $\mathbb{R}^n$, respectively, and let $A = \mathbf{x}\mathbf{y}^T$.
   (a) Show that $\{\mathbf{x}\}$ is a basis for the column space of A and that $\{\mathbf{y}^T\}$ is a basis for the row space of A.
   (b) What is the dimension of $N(A)$?

3. Let A be a matrix. Show that $\text{rank}(A) = \text{rank}(AA^T)$.

4. Let $A \in \mathbb{R}^{m\times n}$, $B \in \mathbb{R}^{n\times r}$, and $C = AB$. Show that
   (a) The column space of C is a subspace of the column space of A.
   (b) The row space of C is a subspace of the row space of B.
   (c) $\text{rank}(C) \leq \min\{\text{rank}(A), \text{rank}(B)\}$.

5. Let A be an $m \times n$ matrix.
   (a) Show that if B is a nonsingular $m \times m$ matrix, then BA and A have the same null space and hence the same rank.
   (b) Show that if C is a nonsingular $n \times n$ matrix, then AC and A have the same rank.

6. True or false. If false, give a counterexample. If true, explain or prove your answer briefly.
   (a) If A is row equivalent to B, then A and B have the same row space.
   (b) If A is a 5×6 matrix of rank 4, then the nullity of A is 1.
   (c) $\mathbb{R}^2$ is a subspace of $\mathbb{R}^4$.
   (d) If $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ are linearly independent, then they span $\mathbb{R}^n$.
   (e) If $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n$ span a vector space V, then they are linearly independent.
   (f) If A is an $m \times n$ matrix, then A and A^T have the same rank.
   (g) If A is an $m \times n$ matrix, then A and A^T have the same nullity.
   (h) If A and B are $n \times n$ matrices that have the same rank, then the rank of AB must equal the rank of BA.